Fault-tolerant Hamiltonian connectivity of the WK-recursive network

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Abstract

A graph G is said to be Hamiltonian-connected if there is a Hamiltonian path between every two distinct nodes of G. Let F denote the set of faulty nodes of G. Then, G is $|F|$-node Hamiltonian-connected if $G - F$ is Hamiltonian-connected. We use $K(d,t)$ to denote a WK-recursive network of level t, each of whose basic modules is a d-node complete graph. Compared with other networks, it is rather difficult to construct a Hamiltonian path between arbitrary two nodes in a faulty WK-recursive network. In this paper, we show that $K(d,t)$ is $(d-4)$-node Hamiltonian-connected. Since the connectivity of $K(d,t)$ is $d-1$, the result is optimal in the worst case.

Keywords: Graph-theoretic interconnection network, WK-recursive, fault-tolerant embedding, Hamiltonian-connected.

1 Introduction

The WK-recursive network, which was originally proposed in [6], represents a family of recursively interconnection networks (networks for short). It has two structural advantages: scalability and constant degree. When a scalable network is expanded, to change node configuration and link connection is not necessary. Additionally, when we add nodes in a network of constant degree, its degree still remains the same. A network with expansibility and equal degree can be easily implemented under rather low cost. A transputer implementation of a 16-node WK-recursive network has been realized at the Hybrid Computing Research Center, Naples, Italy [5]. In this implementation, each node is implemented with the IMS T414 Transputer [14]. Later, this prototype network was further extended to 64 nodes [7]. Recently, much research has been devoted to the WK-recursive networks [4, 5, 7–11, 16–22].

We usually use a graph to represent the topology of an interconnection network. A graph G is a triple consisting of a vertex set $V(G)$, an edge set $E(G)$, and a relation that associates with each edge two vertices called its endpoints [23]. We use (u,v) to denote an edge whose endpoints are u and v. For easy to discuss, we can let $G = V(G) \cup E(G)$. In addition, suppose $S = \{v_0, v_1, ..., v_j\} \subseteq V(G)$. We define a path $P_{v_0,v_j} = \langle v_0, v_1, ..., v_j\rangle \subseteq G$ to be $S \cup \{(v_i, v_{i+1}) | 0 \le i \le j-1\}$. Throughout this study, network and graph, node and vertex, and link and edge, are used interchangeably.

Linear arrays and rings are two fundamental networks for parallel and distributed computation. They are suitable for developing simple algorithms with low communication cost. Many efficient algorithms were designed based on linear arrays and rings for solving a variety of algebraic problems and graph problems [1, 15]. The longest path approach was applied to a practical problem encountered in the on-line optimization of a complex Flexible Manufacturing System [2]. A graph G is called Hamiltonian if there is a Hamiltonian cycle in G, and it is called Hamiltonian-connected [3] if there is a Hamiltonian path between every two distinct nodes of G. A Hamiltonian-connected graph can embed the longest linear array between any two nodes with dilation 1, congestion 1, load 1, and expansion 1.

Since processors or links may fail sometimes, fault-tolerant Hamiltoniancity and fault-tolerant Hamiltonian-connectivity of some networks such as the twisted cube [13], the alternating group graph [?], and the arrangement graph [12] are concerned in many studies. Let $F \subset V(G)$. G is $|F|$-node Hamiltonian (respectively, $|F|$-node Hamiltonian-connected) if $G - F$ is Hamiltonian

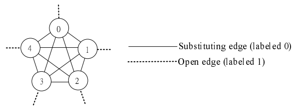

Figure 1: The topology of $K(5,1)$.

(Hamiltonian-connected). If G has connectivity $k+2$ and is k-node Hamiltonian, then it can tolerate a maximal number of node faults while embedding a longest fault-free cycle. If G has connectivity $k+3$ and is k-node Hamiltonian-connected, then it can tolerate a maximal number of node faults while embedding a fault-free Hamiltonian path between any two nodes.

In [11], the author has shown that the WK-recursive network with connectivity $d-1$ is Hamiltonian-connected and $(d-3)$-node Hamiltonian, where $d \geq 4$. In this paper, we improve the previous results to get a stronger statement. We prove that the WK-recursive network with connectivity $d-1$ is $(d-4)$-node Hamiltonian-connected, where $d \geq 4$. That is, the WK-recursive network with connectivity $d-1$ can tolerate $d-4$ node faults while embedding a fault-free Hamiltonian path between any two nodes. Our result is optimal since it can tolerate a maximal number of node faults.

2 Preliminaries

In this section, some important notations and concepts are introduced. The WK-recursive network can be constructed hierarchically by grouping basic modules. The basic module is K_d, a d-node complete graph, where $d > 1$. Throughout this paper, we use $K(d,t)$ to denote a WK-recursive network of level t, each of whose basic modules is K_d, where $d > 1$ and $t \geq 1$. The structures of $K(5,1)$ and $K(5,3)$ are shown in Figure 1 and Figure 2 respectively. We define $K(d,t)$ in terms of a graph as follows.

Definition 1 Each node of $K(d,t)$ is labeled as a t-digit radix d number. A *substituting edge*, labeled as 0, is incident with nodes $a_{t-1}a_{t-2}...a_1b$ and $a_{t-1}a_{t-2}...a_1c$, where $b \neq c$. A *j-flipping edge*, labeled as j, is incident with nodes $a_{t-1}a_{t-2}...a_{j+1}b(c)^j$ and $a_{t-1}a_{t-2}...a_{j+1}c(b)^j$, where $(b)^j$ and $(c)^j$ denote j consecutive b's and c's respectively, and $b \neq c$. For example, the 2-flipping edge connects nodes

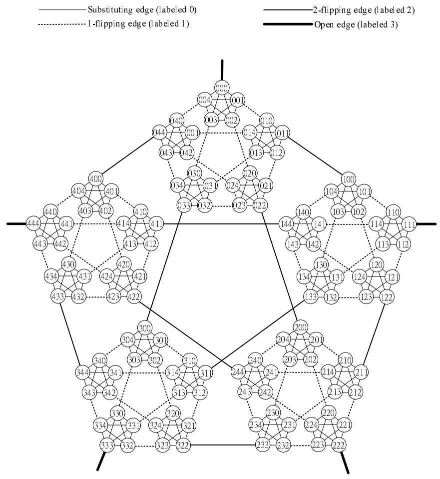

Figure 2: The topology of $K(5,3)$.

033 and 300 in $K(5,3)$ (refer to Figure 2). In addition, an *open edge*, labeled as t, is incident with node $(b)^t$. The open edge is reserved for further expansion; hence, its other end node is unspecified.

Note that $K(d,1)$ is K_d augmented with d open edges (refer to Figure 1). Each node of $K(d,t)$ is incident with $d-1$ substituting edges and one flipping edge (or open edge). The substituting edges are those within basic building blocks, and the j-flipping edges are those connecting two embedded $K(d,j)$'s.

Definition 2 Let $c_{t-1}c_{t-2}\cdots c_m$ be a specific $(t-m)$-digit radix d number. Define $c_{t-1}c_{t-2}\cdots c_m \cdot K(d,m)$ as the subgraph of $K(d,t)$ induced by $\{c_{t-1}c_{t-2}\cdots c_m a_{m-1}\cdots a_1a_0 |\ a_{m-1}\cdots a_1a_0$ is an m-digit radix d number$\}$; that is, $c_{t-1}c_{t-2}...c_m \cdot K(d,m)$ is an embedded $K(d,m)$ with the identifier $c_{t-1}c_{t-2}...c_m$.

In Figure 2, $00 \cdot K(5,1)$ is the subgraph of $K(5,3)$ induced by $\{000, 001, 002, 003, 004\}$. If we regard each $c \cdot K(d,t-1)$, which can be called a *subnetwork* of level $t-1$, as a supernode, then $\{0 \cdot K(d,t-1), 1 \cdot K(d,t-1), 2 \cdot K(d,t-1), ..., (d-1) \cdot K(d,t-1)\}$ forms a complete graph. There is a unique $(t-1)$-flipping edge between two arbitrary supernodes. The $(t-1)$-flipping edge between $a \cdot K(d,t-1)$ and $b \cdot K(d,t-1)$ connects nodes $a(b)^{t-1}$ and $b(a)^{t-1}$.

Definition 3 Node $a_{t-1}a_{t-2}...a_j(b)^j$ is a j-frontier, and is a proper j-frontier if $a_j \neq b$, where $1 \leq j \leq t$.

By definition, a j-frontier is automatically an l-frontier for $1 \leq l \leq j$. A j-frontier is proper means it is not a $(j+1)$-frontier. Both end nodes of a j-flipping edge are proper j-frontiers.

3 Basic idea and some properties

The purpose of this paper is to show that $K(d,t)$ is $(d-4)$-node Hamiltonian-connected, where $d \geq 4$ and $t \geq 1$. In this section, we will give the basic idea of the proof. Additionally, since the proof is rather complicated, some important lemmas are provided in this section. The basic idea is to use induction. Assume that $K(d,t-1)$ is $(d-4)$-node Hamiltonian-connected. We want to prove that $K(d,t)$ is also $(d-4)$-node Hamiltonian-connected. That is, suppose $F \subset V(K(d,t))$ and $|F| \leq d-4$, we want to construct a Hamiltonian path between two arbitrary distinct nodes $A = a_{t-1}a_{t-2}...a_1a_0$ and $B = b_{t-1}b_{t-2}...b_1b_0$ in $K(d,t)-F$.

When A and B are in different subnetworks of level $t-1$, i.e., $a_{t-1} \neq b_{t-1}$, a Hamiltonian path $P_{A,B}$ in $K(d,t)-F$ can be constructed by connecting the $(t-1)$-flipping edges and the Hamiltonian paths for the complement of F in each subnetwork of level $t-1$ (see the proof of Theorem 1 and Figure 4 in the next section). Therefore, we need to find two proper $(t-1)$-frontiers that are not in F for each subnetwork.

When A and B are in the same subnetwork of level $t-1$, i.e., $a_{t-1} = b_{t-1}$, two proper $(t-1)$-frontiers X', Y' in $a_{t-1} \cdot K(d,t-1) - F$ are chosen. Then, two disjoint paths $P_{A,X'}, P_{B,Y'}$ such that $V(P_{A,X'}) \cup V(P_{B,Y'}) = V(a_{t-1} \cdot K(d,t-1) - F)$ are constructed. A Hamiltonian path $P_{A,B}$ in $K(d,t)-F$ can be constructed by connecting $P_{A,X'}$, $P_{B,Y'}$, the $(t-1)$-flipping edges, and the Hamiltonian paths for the complement of F in the remaining subnetworks of level $t-1$ (see the proof of Theorem 1 and Figure 3 in the next section). Therefore, we have to show the existence of paths $P_{A,X'}$, $P_{B,Y'}$ in $K(d,t-1)-F$ and find two proper $(t-1)$-frontiers that are not in F for each subnetwork. We can first prove the lemmas for the existence of paths $P_{A,X'}$, $P_{B,Y'}$ and then use the lemmas to construct a Hamiltonian path for $K(d,t)-F$. However, if $K(d,t-2)$ is $(d-4)$-node Hamiltonian-connected, then constructing $P_{A,X'}$, $P_{B,Y'}$ is easier. Hence, to discuss the construction of a Hamiltonian path and that of the two disjoint paths in $K(d,t)-F$ simultaneously will make the latter task shorter and easier (refer to the proof in Theorem 1 in the next section). Therefore, we combine them together into the main theorem in next section. In the rest of this section, let $F \subset V(K(d,t))$, $t \geq 2$, and $d \geq 5$. For simplicity, we need following definition.

Definition 4 In $K(d,t)$, the sequence $v_0, v_1, \ldots, v_n$ is F-*avoidable* if we have $v_i(v_{i+1})^{t-1}$, $v_{i+1}(v_i)^{t-1} \notin F$ for all $0 \leq i \leq n-1$, where $v_i \neq v_{i+1}$ and $0 \leq v_i, v_{i+1} \leq d-1$.

The following six lemmas, which provide F-avoidable sequences in $K(d,t)$ under various conditions, will be used often in next section.

Lemma 1 Given three arbitrary integers v_0, a', $b' \in \{0,1,2,\ldots,d-1\}$ (v_0, a', b' is not necessarily distinct), if $|F| \leq d-4$, then we can find an F-avoidable sequence $v_0, v_1, \ldots, v_{d-1}, v_0$ in $K(d,t)$ such that $\{v_1, v_{d-1}\} \neq \{a', b'\}$ and $\{v_0, v_1, \ldots, v_{d-1}\} = \{0,1,2,\ldots,d-1\}$.

Proof: We can transform this problem into another problem: the Hamiltonian cycle embedding in $K_d - F_c$, where $F_c \subset E(K_d)$ and $|F_c| \leq d-4$. If we regard each $K(d,t-1)$ as a single supernode, then $K(d,t)$ can be treated as K_d. Suppose $\{0,1,2,\ldots,d-1\}$ is the node set of K_d. Let $i,j \in \{0,1,2,\ldots,d-1\}$, and let i denote a node in K_d corresponding to $i \cdot K(d,t-1)$ in $K(d,t)$. Then, $i(j)^{t-1}$ or $j(i)^{t-1}$ in F is tantamount to $(i,j) \in F_c$. Therefore, $(i,j) \notin F_c$ can assure that $i(j)^{t-1}, j(i)^{t-1} \notin F$. Clearly, we have $|F_c| \leq |F| \leq d-4$. As a result, this lemma is tantamount to the lemma: given three arbitrary integers v_0, a', $b' \in \{0,1,2,\ldots,d-1\}$, if $|F_c| \leq d-4$, then we can find a Hamiltonian cycle $\langle v_0, v_1, ..., v_{d-1}, v_0\rangle$ in $K_d - F_c$ such that $\{v_1, v_{d-1}\} \neq \{a', b'\}$, where $d \geq 5$. The Hamiltonian cycle in $K_d - F_c$ can be constructed by induction on d. Clearly, the lemma holds for $d = 5$. Assume it holds for $d = k \geq 5$. The situation in the case of $d = k+1$ is discussed below. Choose a node $v \in \{0,1,...,k\} - \{a', b', v_0\}$ such that at most one incident edge of v is in F_c. Node v can always be found because $|\{0,1,...,k\} - \{a', b', v_0\}| \geq k-2$ and $|F_c| \leq k-3$.

First, consider that one incident edge of v is in F_c. Then, we have $|(K_{k+1} - \{v\}) \cap F_c| \leq k-4$. With the assumption, there exists a Hamiltonian cycle $\langle v_0, v_1, ..., v_{k-1}, v_0\rangle$ in $(K_{k+1} - \{v\}) - F_c$, where $\{v_1, v_{k-1}\} \neq \{a', b'\}$. Suppose $(v, v_i), (v, v_{(i+1 \mod k)}) \notin F_c$ for some $i \in \{0,1,...,k-1\}$. Then, $(\langle v_0, v_1, ..., v_{k-1}, v_0\rangle -$

$\{(v_i, v_{(i+1 \mod k)})\}) \cup \langle v_i, v, v_{(i+1 \mod k)}\rangle$ is the desired Hamiltonian cycle. Note that the set of the two neighbors of v_0 in the Hamiltonian cycle is still different from $\{a', b'\}$.

Now, consider that there is no incident edge of v in F_c. suppose that $(u, u') \in F_c$. Then, $|(K_{k+1} - \{v\}) \cap (F_c - \{(u, u')\})| \leq k - 4$. With the assumption, there exists a Hamiltonian cycle $\langle v_0, v_1, ..., v_{k-1}, v_0\rangle$ in $(K_{k+1} - \{v\}) - (F_c - \{(u, u')\})$. If (u, u') is in the Hamiltonian cycle, that is, $(u, u') = (v_i, v_{(i+1 \mod k)})$ for some $i \in \{0, 1, ..., k-1\}$, then the construction is the same as the former case. If (u, u') is not in the Hamiltonian cycle, then choose arbitrary $i \in \{0, 1, ..., k-1\}$ and the construction is also the same as the former case. □

Lemma 2 Given two arbitrary distinct integers v_0, $v_{d-1} \in \{0, 1, 2, \ldots, d-1\}$, if $|F| \leq d-4$, then we can find an F-avoidable sequence $v_0, v_1, \ldots, v_{d-1}$ in $K(d, t)$ such that $\{v_0, v_1, \ldots, v_{d-1}\} = \{0, 1, 2, \ldots, d-1\}$.

Proof: Like Lemma 1, this lemma is tantamount to the lemma: there is a Hamiltonian path between two arbitrary nodes in $K_d - F_c$, where $F_c \subset E(K_d)$, $d \geq 5$, and $|F_c| \leq d-4$. Since K_d is $(d-4)$-edge Hamiltonian-connected [10] for $d \geq 5$. The result follows. □

Lemma 3 Given four arbitrary distinct integers v_0, v_{d-1}, e_1, $e_2 \in \{0, 1, 2, \ldots, d-1\}$, if $|F| \leq d-5$ and $v_0(e_1)^{t-1}, v_0(e_2)^{t-1}, e_1(v_0)^{t-1}, e_2(v_0)^{t-1} \notin F$, then we can find an F-avoidable sequence $v_0, v_1, \ldots, v_{d-1}$ in $K(d, t)$ such that $v_1 \in \{e_1, e_2\}$ and $\{v_0, v_1, \ldots, v_{d-1}\} = \{0, 1, 2, \ldots, d-1\}$.

Proof: This lemma is tantamount to the lemma: given four arbitrary distinct integers v_0, v_{d-1}, e_1, $e_2 \in \{0, 1, 2, \ldots, d-1\}$, if $|F_c| \leq d-5$ and $(v_0, e_1), (v_0, e_2) \notin F_c$, then we can find a Hamiltonian path $\langle v_0, v_1, ..., v_{d-1}\rangle$ in $K_d - F_c$ such that $v_1 \in \{e_1, e_2\}$, where $F_c \subset E(K_d)$ and $d \geq 5$. The Hamiltonian path in $K_d - F_c$ can be constructed by induction on d. Clearly, the lemma holds for $d = 5$. Assume it holds for $d = k \geq 5$. The situation in the case of $d = k+1$ is discussed below. Choose a node $v \in \{0, 1, ..., k\} - \{e_1, e_2, v_0, v_{d-1}\}$ such that at most one incident edge of v is in F_c. We can always find v because $|\{0, 1, ..., k\} - \{e_1, e_2, v_0, v_{d-1}\}| \geq k-3$ and $|F_c| \leq k-4$. The rest of the proof is similar to that of Lemma 1. □

Lemma 4 Given three distinct integers a, c, e and two integers a', b' in $\{0, 1, 2, \ldots, d-1\}$, if $|F| \leq d-4$, then we can find two F-avoidable sequences $a, v_1, \ldots, v_m$ and $a, v'_1, \ldots, v'_{d-m-1}$ in $K(d, t)$ for some integer $m \in \{1, 2, \ldots, d-2\}$ such that $\{v_1, v'_1\} \neq \{a', b'\}$, $\{v_m, v'_{d-m-1}\} = \{c, e\}$, and $\{a\} \cup \{v_1, v_2, \ldots, v_m\} \cup \{v'_1, v'_2, \ldots, v'_{d-m-1}\} = \{0, 1, \ldots, d-1\}$, where $d \geq 5$.

Proof: This lemma is tantamount to the lemma: given three distinct integers a, c, e and two integers a', b' in $\{0, 1, 2, \ldots, d-1\}$, if $|F_c| \leq d-4$, then we can find a Hamiltonian path $\langle v'_{d-m-1}, v'_{d-m-2}, \ldots, v'_1, a, v_1, v_2, \ldots, v_m\rangle$ in $K_d - F_c$ for some integer $m \in \{1, 2, \ldots, d-2\}$ such that $\{v_1, v'_1\} \neq \{a', b'\}$, $\{v_m, v'_{d-m-1}\} = \{c, e\}$, and $\{a\} \cup \{v_1, v_2, \ldots, v_m\} \cup \{v'_1, v'_2, \ldots, v'_{d-m-1}\} = \{0, 1, \ldots, d-1\}$, where $d \geq 5$ and $F_c \subset E(K_d)$.

For simplicity, let $\langle u_0, u_1, \ldots, u_j, \ldots, u_{d-1}\rangle = \langle v'_{d-m-1}, v'_{d-m-2}, \ldots, v'_1, a, v_1, v_2, \ldots, v_m\rangle$, where $u_j = a$ for some $j \in \{1, 2, \ldots, d-2\}$. Also, this lemma can be proved by induction on d. Clearly, the lemma holds for $d = 5$. Assume it holds for $d = k \geq 5$. The situation in the case of $d = k+1$ is discussed below. Choose a node $u \in \{0, 1, ..., k\} - \{a, c, e, a', b'\}$ such that at most two incident edges of u is in F_c. Note that u exists because $|\{0, 1, ..., k\} - \{a, c, e, a', b'\}| \geq k-4$ and $|F_c| \leq k-3$. Two cases are considered:

Case 1: At least one incident edge of u is in F_c. Thus, we have $|(K_{k+1} - \{u\}) \cap F_c| \leq k-4$. With the assumption, there exists a Hamiltonian path $\langle u_0, u_1, \ldots, u_j, \ldots, u_{k-1}\rangle$ in $(K_{k+1} - \{u\}) - F_c$, where $u_j = a$ for some $j \in \{1, 2, \ldots, k-2\}$ and $\{a', b'\} \neq \{u_{j-1}, u_{j+1}\}$. First, consider that there are edges $(u, u_i), (u, u_{i+1}) \notin F_c$ for some $i \in \{0, 1, \ldots, k-2\}$. Then $\langle u_0, u_1, \ldots, u_j, \ldots, u_{k-1}\rangle - \{(u_i, u_{i+1})\} \cup \langle u_i, u, u_{i+1}\rangle$ is the desired path. Now, consider that $\{(u, u_i), (u, u_{i+1})\} \cap F_c \neq \emptyset$ for all $i \in \{0, 1, \ldots, k-2\}$. We have $k = 5$ and $\{(u, u_1), (u, u_3)\} = F_c$ since at most two incident edges of u are in F_c. Moreover, we have $(K_{k+1} - \{u\}) - F_c = K_k$. If $a = u_2$, then $\langle u_0, u, u_2, u_1, u_3, u_4\rangle$ is the desired path. If $a = u_1$ and $\{u_2, u_3\} = \{a', b'\}$, then $\langle u_0, u_1, u_3, u_2, u, u_4\rangle$ is the desired path. If $a = u_1$ and $\{u_2, u_3\} \neq \{a', b'\}$, then $\langle u_0, u, u_2, u_1, u_3, u_4\rangle$ is the desired path. In addition, the construction of the case when

$a = u_3$ is similar to that of the case when $a = u_1$.

Case 2: No incident edge of u is in F_c. The proof is similar to that of Lemma 1. □

Lemma 5 Given four arbitrary distinct integers $a, b, c, e \in \{0, 1, 2, \ldots, d-1\}$, if $|F| \leq d-5$, then we can find two F-avoidable sequences $v_1, v_2, \ldots, v_m$ and $v'_1, v'_2, \ldots, v'_{d-m}$ in $K(d,t)$ for some integer $m \in \{2, 3, \ldots, d-2\}$ such that $v_1 = a$, $v'_1 = b$, $\{v_m, v'_{d-m}\} = \{c, e\}$, and $\{v_1, v_2, \ldots, v_m\} \cup \{v'_1, v'_2, \ldots, v'_{d-m}\} = \{0, 1, \ldots, d-1\}$.

Proof: This lemma is tantamount to the lemma: given four arbitrary distinct integers $a, b, c, e \in \{0, 1, 2, \ldots, d-1\}$, if $|F_c| \leq d-5$, then we can find two paths $\langle v_1, v_2, \ldots, v_m\rangle$ and $\langle v'_1, v'_2, \ldots, v'_{d-m}\rangle$ in $k_d - F_c$ for some integer $m \in \{2, 3, \ldots, d-2\}$ such that $a = v_1$, $b = v'_1$, $\{c, e\} = \{v_m, v'_{d-m}\}$, and $\{v_1, v_2, \ldots, v_m\} \cup \{v'_1, v'_2, \ldots, v'_{d-m}\} = \{0, 1, \ldots, d-1\}$, where $F_c \subset E(K_d)$ and $d \geq 5$. Also, this lemma can be proved by induction on d. Clearly, the lemma holds for $d = 5$. Assume it holds for $d = k \geq 5$. The situation in the case of $d = k+1$ is discussed below. Choose a node $v \in \{0, 1, ..., k\} - \{a, b, c, e\}$ such that at most one incident edge of v is in F_c. Node v can always be found because $|\{0, 1, ..., k\} - \{a, b, c, e\}| \leq k-3$ and $|F_c| \leq k-4$. The rest of the proof is similar to that of Lemma 1. □

Lemma 6 Given two distinct integers a, b and an integer a' in $\{0, 1, \ldots, d-1\}$, if $|F| \leq d-4$, then we can find two F-avoidable sequences $v_1, v_2, \ldots, v_m$ and $v'_1, v'_2, \ldots, v'_{d-m}$ in $K(d,t)$ for some integer $m \in \{1, 2, \ldots, d-1\}$ such that $v_1 = a$, $v'_1 = b$, $a' \notin \{v_m, v'_{d-m}\}$, and $\{v_1, v_2, \ldots, v_m\} \cup \{v'_1, v'_2, \ldots, v'_{d-m}\} = \{0, 1, \ldots, d-1\}$.

Proof: This lemma is tantamount to the lemma: given two distinct integers a, b and an integer a' in $\{0, 1, \ldots, d-1\}$, if $|F_c| \leq d-4$, then we can find two paths $\langle v_1, v_2, \ldots, v_m\rangle$ and $\langle v'_1, v'_2, \ldots, v'_{d-m}\rangle$ in $K_d - F_c$ for some integer $m \in \{1, 2, \ldots, d-1\}$ such that $a = v_1$, $b = v'_1$, $a' \notin \{v_m, v'_{d-m}\}$, and $\{v_1, v_2, \ldots, v_m\} \cup \{v'_1, v'_2, \ldots, v'_{d-m}\} = \{0, 1, \ldots, d-1\}$, where $F_c \subset E(K_d)$ and $d \geq 5$. The rest of the proof is similar to that of Lemma 1. □

4 Main theorem

In this section, we will prove the main theorem. In [11], the author has shown that $K(d,t)$ is Hamiltonian-connected, where $d \geq 4$ and $t \geq 1$. We have only to consider the situation when $d \geq 5$ since $d-4 = 0$ if $d = 4$. For simplicity, we would like to combine two lemmas into the main theorem as follows.

Theorem 1 Suppose $F \subset V(K(d,t))$. Let $A = a_{t-1}a_{t-2}...a_1a_0$ and $B = b_{t-1}b_{t-2}...b_1b_0$ be two distinct nodes in $K(d,t) - F$, where $d \geq 5$.

1). If $|F| \leq d-4$, then there is a Hamiltonian path $P_{A,B}$ in $K(d,t) - F$.

2). Given two distinct integers $c, e \in \{0, 1, ..., d-1\}$ such that $\{c, e\} \neq \{a_{t-1}, b_{t-1}\}$ and $(c)^t$, $(e)^t \notin F$, if $|F| \leq d-5$, then there exist two disjoint paths $P_{A,X}$ and $P_{B,Y}$ such that $V(P_{A,X}) \cup V(P_{B,Y}) = V(K(d,t) - F)$, where $\{X, Y\} = \{(c)^t, (e)^t\}$.

3). Given an integer $a' \in \{0, 1, ..., d-1\}$, if $|F| = d-4$, then we can find two distinct integers $c, e \in \{0, 1, ..., d-1\} - \{a'\}$ and construct two disjoint paths $P_{A,X}$ and $P_{B,Y}$ such that $V(P_{A,X}) \cup V(P_{B,Y}) = V(K(d,t) - F)$, where $\{c, e\} \neq \{a_{t-1}, b_{t-1}\}$ and $X = (c)^t, Y = (e)^t$.

Proof: We proceed by induction on t. Remember that $K(d,1)$ is K_d augmented with d open edges. Ignore open edges, we can just regard $K(d,1)$ as K_d. The theorem holds for $t = 1$ since $K(d,t) - F = K_d - F$ is still a complete graph. Assume the theorem holds for $t = k \geq 1$. The situation in the case of $t = k+1$ is discussed below. First, we will prove Part 1. Two cases are considered:

Case 1: $a_k = b_k$, that is, A and B are in the same subnetwork of level k. Let $v_0 = a_k$. Two cases are further considered:

Case 1.1: $|F \cap a_k \cdot K(d,k)| \leq d-5$. According to Lemma 1, we can find an F-avoidable sequence $v_0, v_1, \ldots, v_{d-1}, v_0$ in $K(d, k+1)$ such that $\{v_1, v_{d-1}\} \neq \{a_{k-1}, b_{k-1}\}$ and $\{v_0, v_1, \ldots, v_{d-1}\} = \{0, 1, 2, \ldots, d-1\}$. With the assumption of Part 2, there exist two disjoint paths $P_{A,X'}$ and $P_{B,Y'}$ such that $V(P_{A,X'}) \cup V(P_{B,Y'}) = V(a_k \cdot K(d,t) - F)$, where $\{X', Y'\} = \{v_0(v_1)^k, v_0(v_{d-1})^k\}$. In addition, with the assumption of Part 1, there exists a Hamiltonian path for $v_i \cdot K(d,k) - F$ between two arbitrary distinct nodes in

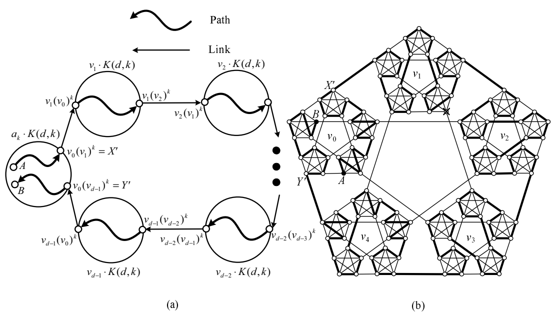

Figure 3: (a) The construction and (b) an example of a Hamiltonian path $P_{A,B}$ for $K(d, k + 1)$ when $a_k = b_k$.

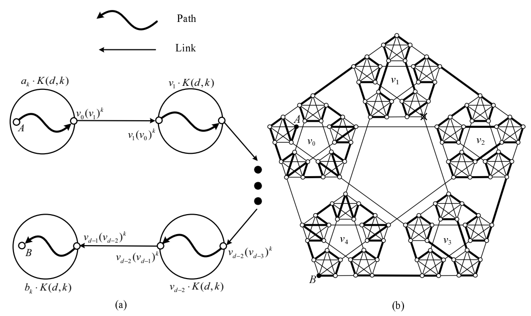

Figure 4: (a) The construction and (b) an example of a Hamiltonian path $P_{A,B}$ for $K(d, k + 1)$ when $a_k \neq b_k$.

$v_i \cdot K(d, k) - F$ for each $i \in \{1, 2, ..., d - 1\}$. Without loss of generality, assume that $X' = v_0(v_1)^k$ and $Y' = v_0(v_{d-1})^k$. The desired Hamiltonian path $P_{A,B}$ can be constructed as shown in Figure 3.

Case 1.2: $|F \cap a_k \cdot K(d, k)| = d - 4$. With the assumption of Part 3, we can find two distinct integers $v_1, v_{d-1} \in \{0, 1, ..., d-1\} - \{v_0\}$ and construct two disjoint paths $P_{A,X'}$ and $P_{B,Y'}$ such that $V(P_{A,X'}) \cup V(P_{B,Y'}) = V(a_k \cdot (d, k) - F)$, where $\{v_1, v_{d-1}\} \neq \{a_{k-1}, b_{k-1}\}$ and $X' = v_0(v_1)^k, Y' = v_0(v_{d-1})^k$. Additionally, let $\{v_0, v_1, \ldots, v_{d-1}\} = \{0, 1, 2, \ldots, d - 1\}$. Clearly, because $|F \cap a_k \cdot K(d, k)| = d - 4$, we have $F \cap i \cdot K(d, k) = \emptyset$ for all $i \in \{0, 1, 2, \ldots, d - 1\} - \{v_0\}$. Thus, $v_0, v_1, \ldots, v_{d-1}, v_0$ is certainly an F-avoidable sequence in $K(d, k+1)$. The desired Hamiltonian path $P_{A,B}$ can be constructed as shown in Figure 3.

Case 2: $a_k \neq b_k$, that is, A and B are not in the same subnetwork of level k. Let $v_0 = a_k$, $v_{d-1} = b_k$. According to Lemma 2, we can find a F-avoidable sequence $v_0, v_1, \ldots, v_{d-1}$ in $K(d, k + 1)$ such that $\{v_0, v_1, \ldots, v_{d-1}\} = \{0, 1, 2, \ldots, d - 1\}$. The desired Hamiltonian path $P_{A,B}$ can be constructed as shown in Figure 4.

Now, we will prove Part 2. That is, given two distinct integers $c, e \in \{0, 1, ..., d - 1\}$ such that $\{c, e\} \neq \{a_k, b_k\}$ and $(c)^{k+1}$, $(e)^{k+1} \notin F$, if $|F| \leq d-5$, then there exist two disjoint paths $P_{A,X}$ and $P_{B,Y}$ such that $V(P_{A,X}) \cup V(P_{B,Y}) = V(K(d, k+1) - F)$, where $\{X, Y\} = \{(c)^{k+1}, (e)^{k+1}\}$. Two cases are considered:

Case 1: $a_k = b_k$. Two cases are further considered:

Case 1.1: $a_k \in \{c, e\}$, that is, exactly one of $(c)^{k+1}$ or $(e)^{k+1}$ is in $a_k \cdot K(d, k)$. Without loss of generality, suppose $a_k = c$. We choose two distinct integers e_1 and e_2 from $\{0, 1, 2, \ldots d - 1\} - \{e, a_{k-1}, b_{k-1}\}$ (respectively, from $\{0, 1, 2, \ldots d - 1\} - \{c, e\}$) if $c \in \{a_{k-1}, b_{k-1}\}$ (respectively, if $c \notin \{a_{k-1}, b_{k-1}\}$) such that $a_k(e_1)^k$, $a_k(e_2)^k$, $e_1(a_k)^k$, $e_2(a_k)^k \notin F$. Note that e_1 and e_2 exists since $|F| \leq d - 5$ and $d - (d - 5) - 3 \geq 2$. Let $v_0 = a_k$, $v_{d-1} = e$. According to Lemma 3, we can find an F-avoidable sequence $v_0, v_1, \ldots, v_{d-1}$ in $K(d, k + 1)$ such that $v_1 \in \{e_1, e_2\}$ and $\{v_0, v_1, \ldots, v_{d-1}\} = \{0, 1, 2, \ldots, d-1\}$. In addition, we have $\{c, v_1\} \neq \{a_{k-1}, b_{k-1}\}$ since $v_1 \notin \{a_{k-1}, b_{k-1}\}$ if $c \in \{a_{k-1}, b_{k-1}\}$. Moreover, remember that $a_k(c)^k, a_k(v_1)^k \notin F$. With the assumption of Part 2, there exist two disjoint paths $P_{A,X'}$ and $P_{B,Y'}$ such that $V(P_{A,X'}) \cup V(P_{B,Y'}) = V(a_k \cdot K(d, t) - F)$, where $\{X', Y'\} = \{a_k \cdot (c)^k, a_k \cdot (v_1)^k\}$. Without loss of generality, suppose $X = X' = a_k \cdot (c)^k$, the desired paths $P_{A,X}$ and $P_{B,Y}$ can be constructed as shown in Figure 5.

Case 1.2: $a_k \notin \{c, e\}$, that is, neither $(c)^{k+1}$ nor $(e)^{k+1}$ are in $a_k \cdot K(d, k)$. Note that a_k, c, and e are distinct. According to Lemma 4, we can find two F-avoidable sequences $a_k, v_1, v_2, \ldots, v_m$ and $a_k, v'_1, v'_2, \ldots, v'_{d-m-1}$ in $K(d, k + 1)$ for some integer $m \in \{1, 2, \ldots, d - 2\}$ such that $\{v_1, v'_1\} \neq \{a_{k-1}, b_{k-1}\}$, $\{v_m, v'_{d-m-1}\} = \{c, e\}$, and $\{a_k\} \cup \{v_1, v_2, \ldots, v_m\} \cup \{v'_1, v'_2, \ldots, v'_{d-m-1}\} = \{0, 1, \ldots, d - 1\}$.

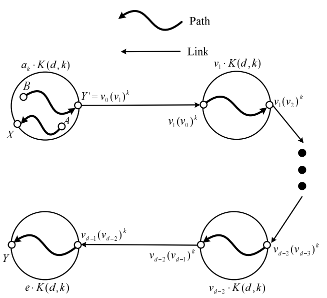

Figure 5: The construction of two disjoint paths $P_{A,X}$ and $P_{B,Y}$ when $a_k = b_k = c$.

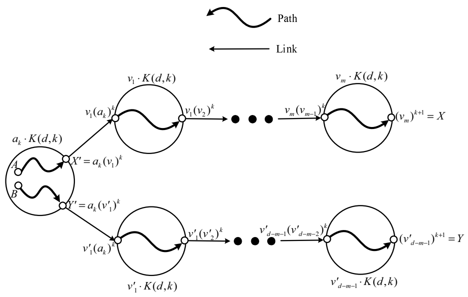

Figure 6: The construction of two disjoint paths $P_{A,X}$ and $P_{B,Y}$ when $c \neq a_k = b_k \neq e$.

With the assumption of Part 2, there exist two disjoint paths $P_{A,X'}$ and $P_{B,Y'}$ such that $V(P_{A,X'}) \cup V(P_{B,Y'}) = V(a_k \cdot K(d,k) - F)$, where $\{X', Y'\} = \{a_k(v_1)^k, a_k(v_1')^k\}$. Without loss of generality, suppose $X' = a_k(v_1)^k$. The desired paths $P_{A,X}$ and $P_{B,Y}$ can be constructed as shown in Figure 6.

Case 2: $a_k \neq b_k$. Two cases are further considered:

Case 2.1: $\{c, e\} \cap \{a_k, b_k\} \neq \emptyset$. Without loss of generality, suppose $a_k = c$, that is, $(c)^{k+1}$ and A are in the same subnetwork of level k (then $b_k \notin \{c, e\}$ since $\{c, e\} \neq \{a_k, b_k\}$). In addition, note that a_k, b_k, e are distinct. According to Lemma 4, we can find two F-avoidable sequences $a_k, v_1, v_2, \ldots, v_m$ and $a_k, v_1', v_2', \ldots, v_{d-m-1}'$ in $K(d, k+1)$ for some integer $m \in \{1, 2, \ldots, d-2\}$ such that $\{v_1, v_1'\} \neq \{a_{k-1}, c\}$, $\{v_m, v_{d-m-1}'\} = \{b_k, e\}$ and $\{a_k\} \cup \{v_1, v_2, \ldots, v_m\} \cup \{v_1', v_2', \ldots, v_{d-m-1}'\} = \{0, 1, \ldots, d-1\}$. Assume that $v_m = b_k$ and $v_{d-m-1}' = e$. (The construction is similar when $v_m = e$ and $v_{d-m-1}' = b_k$.)

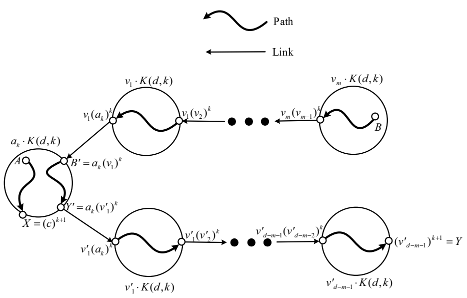

Figure 7: The construction of two disjoint paths paths $P_{A,X}$ and $P_{B,Y}$ when $a_k \neq b_k$, $a_k = c$ $X' = a_k(c)^k$, and $Y' = a_k(v_1')^k$.

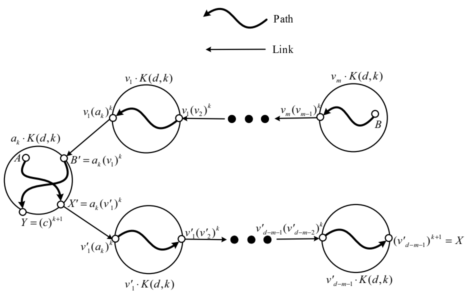

Figure 8: The construction of two disjoint paths paths $P_{A,X}$ and $P_{B,Y}$ when $a_k \neq b_k$, $a_k = c$ $Y' = a_k(c)^k$, and $X' = a_k(v_1')^k$.

Since $a_k = c \neq v_1$ and $v_1' \neq v_1$, we have $\{c, v_1'\} \neq \{a_{k-1}, v_1\}$. Let $B' = a_k(v_1)^k$. With the assumption of Part 2, there exist two disjoint paths $P_{A,X'}$ and $P_{B',Y'}$ such that $V(P_{A,X'}) \cup V(P_{B',Y'}) = V(a_k \cdot K(d,k) - F)$, where $\{X', Y'\} = \{a_k(c)^k, a_k(v_1')^k\}$. If $X' = a_k(c)^k$ and $Y' = a_k(v_1')^k$, then let $X = (c)^{k+1} = a_k(c)^k = X'$ and $Y = (e)^{k+1}$, and the desired paths $P_{A,X}$ and $P_{B,Y}$ can be constructed as shown in Figure 7. If $Y' = a_k(c)^k$ and $X' = a_k(v_1')^k$, then let $Y = a_k(c)^k$ and $X = (e)^{k+1}$, and the desired paths $P_{A,X}$ and $P_{B,Y}$ can be constructed as shown in Figure 8.

Case 2.2: $\{c, e\} \cap \{a_k, b_k\} = \emptyset$. Thus, a_k, b_k, c, and e are distinct. According to Lemma 5, we can find two F-avoidable sequences $v_1, v_2, \ldots, v_m$ and $v_1', v_2', \ldots, v_{d-m}'$ in $K(d, k+1)$ for some integer $m \in \{2, 3, \ldots, d-2\}$ such that $v_1 = a_k$, $v_1' = b_k$, $\{v_m, v_{d-m}'\} = \{c, e\}$, and $\{v_1, v_2, \ldots, v_m\} \cup \{v_1', v_2', \ldots, v_{d-m}'\} = \{0, 1, \ldots, d-1\}$. Without loss of generality, suppose that $v_m = c$ and $v_{d-m}' = e$. The de-

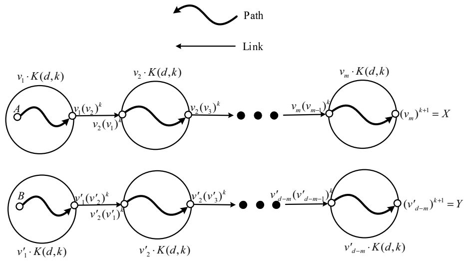

Figure 9: The construction of two disjoint paths $P_{A,X}$ and $P_{B,Y}$ when $\{c,e\} \cap \{a_k, b_k\} = \emptyset$.

sired paths $P_{A,X}$ and $P_{B,Y}$ can be constructed as shown in Figure 9.

Finally, we will prove Part 3. That is, given an integer $a' \in \{0,1,...,d-1\}$, if $|F| = d-4$, then we can find two distinct integers $c,e \in \{0,1,...,d-1\} - \{a'\}$ and construct two disjoint paths $P_{A,X}$ and $P_{B,Y}$ such that $V(P_{A,X}) \cup V(P_{B,Y}) = V(K(d,k+1) - F)$, where $\{c,e\} \neq \{a_k, b_k\}$ and $X = (c)^{k+1}, Y = (e)^{k+1}$. Two cases will be considered:

Case 1: $a_k = b_k$. Two cases are further considered:

Case 1.1: $|F \cap a_k \cdot K(d,k)| \leq d-5$. Since $|F| \leq d-4$, we can find two distinct integers $u_1, u_2 \in \{0,1,...,d-1\} - \{a_k, a'\}$ such that $(u_1)^{k+1}, (u_2)^{k+1} \in K(d,k+1) - F$. Note that a_k, u_1, u_2 are distinct. According to Lemma 4, we can find two F-avoidable sequences $a_k, v_1, v_2, \ldots, v_m$ and $a_k, v'_1, v'_2, \ldots, v'_{d-m-1}$ in $K(d,k+1)$ for some integer $m \in \{1,2,\ldots,d-2\}$ such that $\{v_1, v'_1\} \neq \{a_{k-1}, b_{k-1}\}$, $\{v_m, v'_{d-m-1}\} = \{u_1, u_2\}$, and $\{a_k\} \cup \{v_1, v_2, \ldots, v_m\} \cup \{v'_1, v'_2, \ldots, v'_{d-m-1}\} = \{0,1,\ldots,d-1\}$. Since $|F \cap a_k \cdot K(d,k)| \leq d-5$, with the assumption of Part 2, there exist two disjoint paths $P_{A,X'}$ and $P_{B,Y'}$ such that $V(P_{A,X'}) \cup V(P_{B,Y'}) = V(a_k \cdot K(d,k) - F)$, where $\{X', Y'\} = \{a_k(v_1)^k, a_k(v'_1)^k\}$. Without loss of generality, suppose that $X' = a_k(v_1)^k$. Let $c = v_m, e = v'_{d-m-1}$, $X = (c)^{k+1}$, and $Y = (e)^{k+1}$. The desired paths $P_{A,X}$ and $P_{B,Y}$ can be constructed as shown in Figure 6.

Case 1.2: $|F \cap a_k \cdot K(d,k)| = d-4$. Clearly, $(K(d,k+1) - a_k \cdot K(d,k)) \cap F = \emptyset$. With the assumption of Part 3, we can find $c', e' \in \{0,1,...,d-1\} - \{a_k\}$ and construct two disjoint paths $P_{A,X'}$ and $P_{B,Y'}$ such that $V(P_{A,X'}) \cup V(P_{B,Y'}) = V(a_k \cdot K(d,k) - F)$, where $\{c', e'\} \neq \{a_{k-1}, b_{k-1}\}$ and $X' = a_k(c')^k, Y' = a_k(e')^k$. Since $(K(d,k+1) - a_k \cdot K(d,k)) \cap F = \emptyset$, we can easily find two F-avoidable sequences $a_k, v_1, \ldots, v_m$ and $a_k, v'_1, \ldots, v'_{d-m-1}$ in $K(d,k+1)$ for some integer $m \in \{1,2,\ldots,d-2\}$ such that $a' \notin \{v_m, v'_{d-m-1}\}$, and $\{a_k\} \cup \{v_1, \ldots, v_m\} \cup \{v'_1, v'_2, \ldots, v'_{d-m-1}\} = \{0,1,\ldots,d-1\}$. Let $c = v_m, e = v'_{d-m-1}$, $X = (c)^{k+1}$, and $Y = (e)^{k+1}$. The desired paths $P_{A,X}$ and $P_{B,Y}$ can be constructed as shown in Figure 6.

Case 2: $a_k \neq b_k$. According to Lemma 6, we can find two F-avoidable sequences $v_1, v_2, \ldots, v_m$ and $v'_1, v'_2, \ldots, v'_{d-m}$ in $K(d,k+1)$ for some integer $m \in \{1,2,\ldots,d-2\}$ such that $v_1 = a_k$, $v'_1 = b_k$, $a' \notin \{v_m, v'_{d-m}\}$, and $\{v_1, v_2, \ldots, v_m\} \cup \{v'_1, v'_2, \ldots, v'_{d-m}\} = \{0,1,\ldots,d-1\}$. Let $c = v_m, e = v'_{d-m-1}$, $X = (c)^{k+1}$, and $Y = (e)^{k+1}$. Note that it is possible that $a_k = c$ or $b_k = e$ but not both. Thus, $\{c,e\} \neq \{a_k, b_k\}$. The desired paths $P_{A,X}$ and $P_{B,Y}$ can be constructed as shown in Figure 9. □

5 Discussion and conclusion

In this paper, we show that $K(d,t)$ is $(d-4)$-node Hamiltonian-connected, where $d \geq 4$ and $t \geq 1$. That is, there is a Hamiltonian path between every two distinct nodes of the network that remains after removing any $d-4$ nodes from $K(d,t)$. Our results reveal that the WK-recursive network can tolerate $d-4$ node faults while embedding the longest linear array between any two distinct nonfaulty nodes with dilation, congestion, load, and expansion equal to one. Our result is optimal since the connectivity of $K(d,t)$ is $d-1$.

According to our proof, it is not difficult to design an algorithm containing three indirect recursive functions in order to construct a Hamiltonian path between two arbitrary distinct nonfaulty nodes in the network that remains after removing $d-4$ node faults from $K(d,t)$.

Many nodes in the WK-recursive network can not connect to other subnetworks directly, whereas all nodes in most of networks such as the twisted cube and the arrangement graph can do. Because of this situation, Part 2 and Part 3 of Theorem 1, which are rather sophisticated, can not be avoided. Refer to Figure 10, if all nodes in $a_k \cdot K(d,k)$ can connect to other subnetworks directly, then

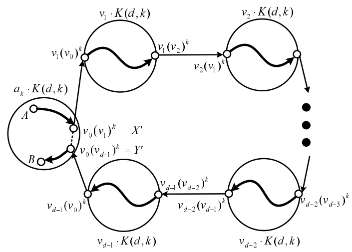

Figure 10: The construction a Hamiltonian path $P_{A,B}$ for $K(d, k+1)$ when $a_k = b_k$ if each node in $a_k \cdot K(d,k)$ can connect to other subnetworks directly.

two disjoint paths in $a_k \cdot K(d,k) - F$ can be constructed by splitting up the Hamiltonain path for $a_k \cdot K(d,k) - F$. And, the Hamiltonian path $P_{A,B}$ can be constructed without Part 2 and Part 3. Therefore, the fault-tolerant Hamiltonian connectivity of the WK-recursive network is more complicated to that of other networks naturally.

Acknowledgments

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